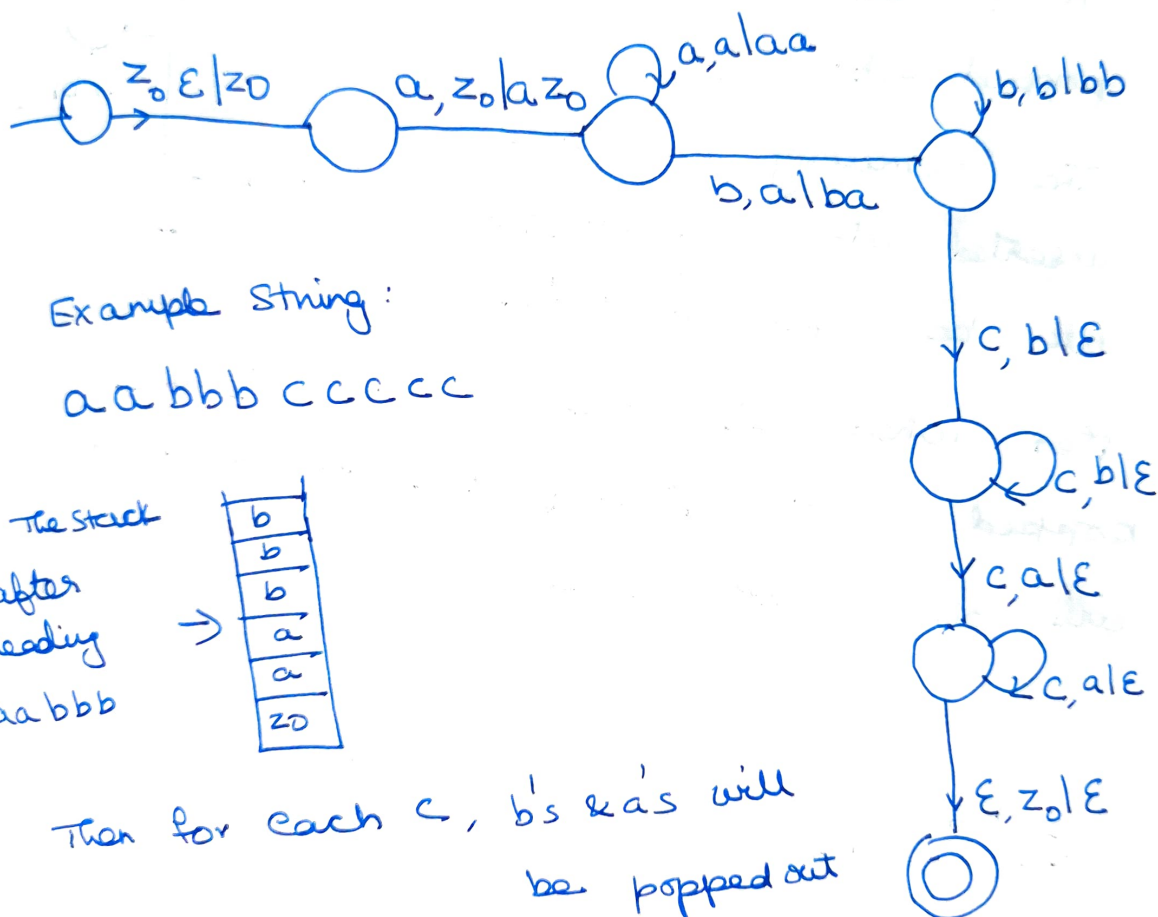


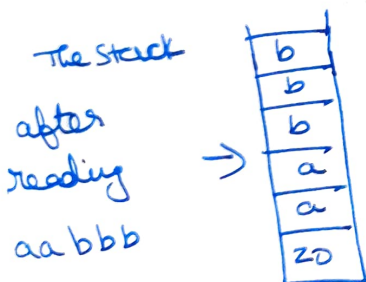
$$1) \{ a^n b^m c^{n+m} \mid n \geq 0, m \geq 0 \}$$

Push all a's & b's inside stack. When encountering c's (which should be equal to the total of a's & b's) Pop b's & then a's.

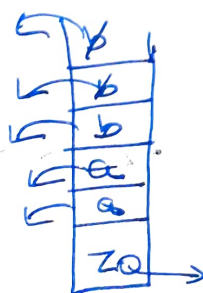


Example String:

a a b b b c c c c c



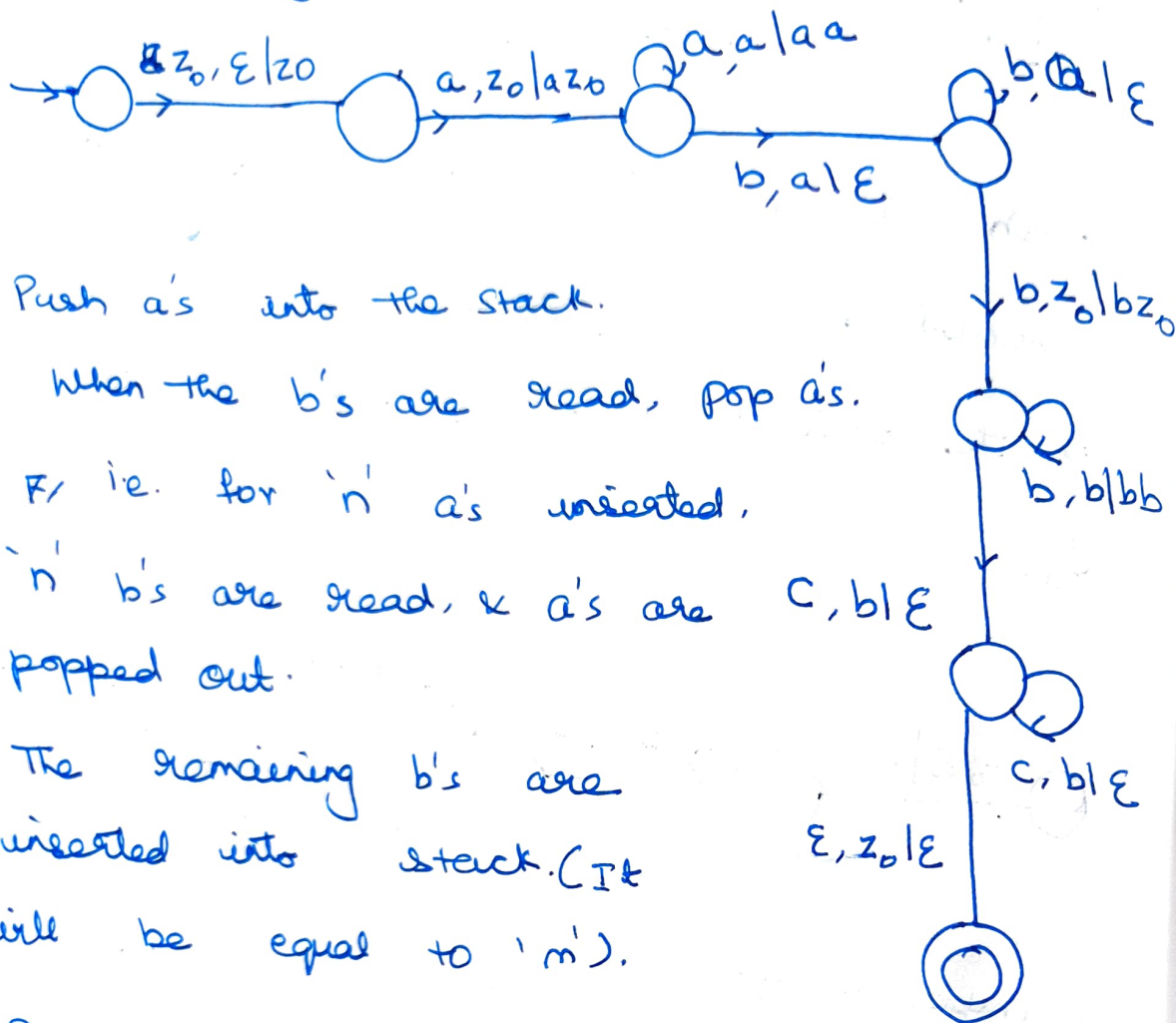
Then for each c, b's & a's will be popped out one by one



finally this will be popped out.

2) $a^n b^{n+m} c^m$

Example string: $a a b b b b b c c c$



Push a's into the stack.

When the b's are read, pop a's.

E.g. ie. for 'n' a's inserted,

'n' b's are read, & a's are popped out.

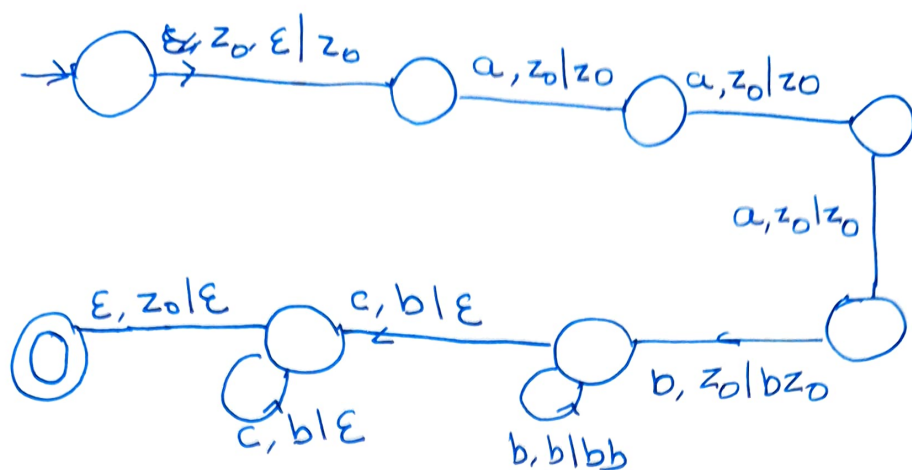
The remaining b's are inserted into stack. (It will be equal to 'm').

Then when c's are read, b's are popped out. Finally the stack symbol will be left out.

3) $a^3 b^n c^n \mid n \geq 0$

Example: $a a a b b b b \rightarrow$ where $n = 2$.





First insert stack symbol, then 3 a's are read. There is no push or pop for the a's. After processing 3 a's, then the b's read are inserted into stack. Then when c's are read, each b is popped out. Finally the stack symbol is left out.

$$4) L = \{w : n_a(w) = n_b(w) + 1\}$$

The number of a's should be one greater than number of b's.

Ex: a a a b b

